Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Non-Contact Upgrade of LS SWIR SDK for Reliable Spectral Measurement at a Distance of ≥10 cm**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electronics / Semiconductor Systems		*AI / Data Processing*	
	Medium Category	Small Category	Medium Category	Small Category
	Optical & Sensor Systems	*SWIR Spectral Sensing Modules and Measurement Systems*	*Signal Processing & Machine Learning*	*Distance-Robust Spectral Calibration and Classification*
Concept of Proposed Technology	**Core Concept** Compact SWIR (Short-wave Infrared) spectral sensing systems based on contact or near-contact measurement have demonstrated high effectiveness for portable material analysis with minimal system complexity. Building on this foundation, the proposed technology extends SWIR sensing into a robust non-contact system through integrated sensor–optics co-design and AI-based compensation, enabling scalable applications such as conveyor-based sorting, robotic inspection, and environmental monitoring (Fig. 1). **Key Functions and Potential Applications** The proposed system enables high-performance non-contact SWIR spectral sensing at standoff distances of 10 cm or greater, providing accurate and reliable measurements under optically uncontrolled industrial conditions. Through integrated sensor–optics design and AI-based compensation, it maintains system reliability, compactness, and cost efficiency while supporting continuous and automated operation. This technology can be applied to conveyor-based material sorting, robotic inspection, automated quality control, industrial process monitoring, environmental			

sensing, agricultural analysis, and government inspection systems, enabling scalable deployment across large-scale manufacturing and field environments.

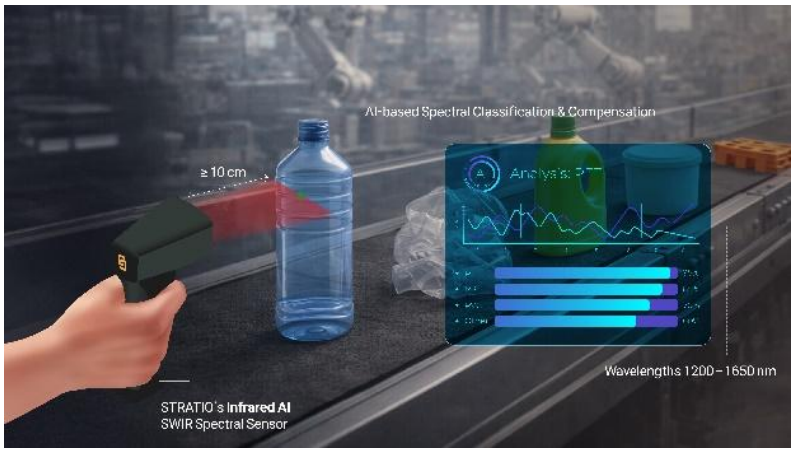

Figure 1. Conceptual Image of an optical system for long-distance, non-contact measurement

Rationale for Joint R&D Project

Critical Technical Barriers

SWIR sensing has long been used in spectroscopy and material analysis due to its ability to distinguish intrinsic material characteristics. However, most conventional SWIR sensors are based on compound semiconductors such as indium gallium arsenide (InGaAs), whose material and fabrication costs have historically limited practical deployment to laboratories, defense applications, and other cost-insensitive domains.

Recent commercialization of germanium (Ge)-based SWIR sensor architectures, including gate-controlled modulated device (GCMD)–type approaches [1], has demonstrated that comparable SWIR absorption performance can be achieved using silicon-compatible materials. This enables wafer-scale integration and significantly improved cost scalability, establishing a realistic foundation for expanding SWIR-based material intelligence into broader industrial and commercial environments. While such sensor platforms have proven effective in contact-based configurations, extending them to reliable non-contact operation remains a critical system-level challenge addressed by this project.Contact-based and near-contact spectral sensing systems are constrained in automated and large-scale industrial environments due to limited throughput, contamination risks, and mechanical coupling requirements. These physical contact dependencies reduce system scalability and reliability in continuous operation settings, highlighting the need for stable and accurate non-contact measurement solutions.

The transition from contact-based to non-contact SWIR spectral sensing introduces fundamental optical and photometric challenges that cannot be addressed through incremental sensor or algorithm optimization alone. As the sensor-to-target distance increases to 10 cm or greater, signal intensity decreases by nearly two orders of magnitude due to inverse-square-law attenuation. This severe photometric decay is further compounded by increased susceptibility to ambient illumination and stray infrared radiation in optically uncontrolled industrial environments. As a result, signal-to-noise ratio (SNR) degradation significantly limits spectral fidelity and classification accuracy, creating a fundamental performance barrier for reliable non-contact operation.

Furthermore, insufficient photon collection at the optical front end cannot be compensated through software or AI-based post-processing. While the compact, structured SWIR sensor architecture enables high-resolution spectral acquisition, its small active area imposes strict requirements on optical focusing and photon delivery efficiency. Reliable non-contact performance is therefore constrained by the ability to collect, control, and deliver sufficient photons under distance-induced attenuation. Overcoming these limitations requires a system-level redesign centered on high-efficiency optical collection, precise alignment, and joint optimization of optical, sensor, and signal-processing subsystems, rather than isolated component-level improvements.

Expanding the Operational Envelope: From "Touch" to "Scan"

The proposed system is built upon a GCMD-based SWIR sensor architecture that enables compact and structured spectral acquisition through monolithic wafer-level integration of bandpass filter (BPF) arrays. This integration allows the realization of a highly compact, mechanically robust spectral engine capable of resolving material-specific signatures across the 1200–1650 nm wavelength range (Fig. 2 & 3). By combining spectral filtering and sensing elements at the wafer scale, the platform provides a stable and manufacturable foundation that supports the transition from contact-based operation to distance-based spectral scanning.

From a non-contact sensing perspective, the small active area and structured detection characteristics of the sensor enable aggressive optical focusing and precise photon delivery, which are essential for extending reliable operation from "touch" measurements to "scan"-based sensing at practical standoff distances. However, under distance-induced attenuation, overall system performance is fundamentally constrained by photon collection efficiency and optical coupling quality. The proposed project addresses this limitation through advanced optical system design and co-optimization strategies that tightly integrate sensing, illumination, and alignment subsystems. By establishing high-efficiency photon management as a core system capability, this approach expands the operational envelope of compact SWIR spectroscopy from controlled contact environments to dynamic, non-contact industrial settings.

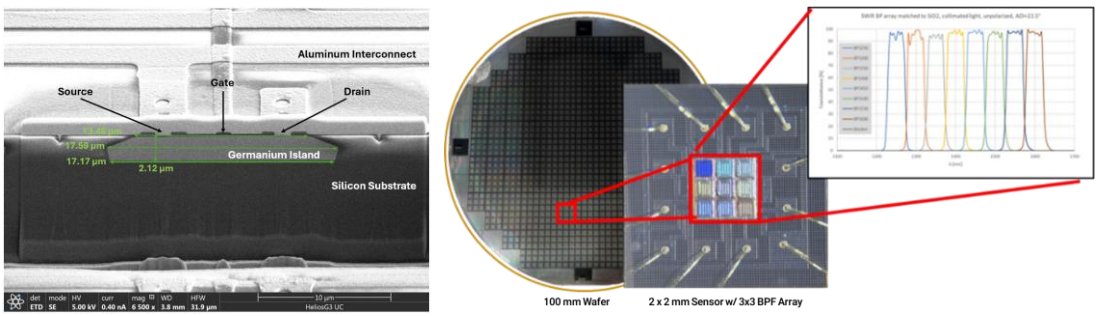

Figure 2. Cross-Sectional SEM Image of Ge SWIR sensor array (left) and SWIR spectral sensor in LS SWIR SDK fabricated using wafer-scale integration of Ge Sensor & BPF array (right)

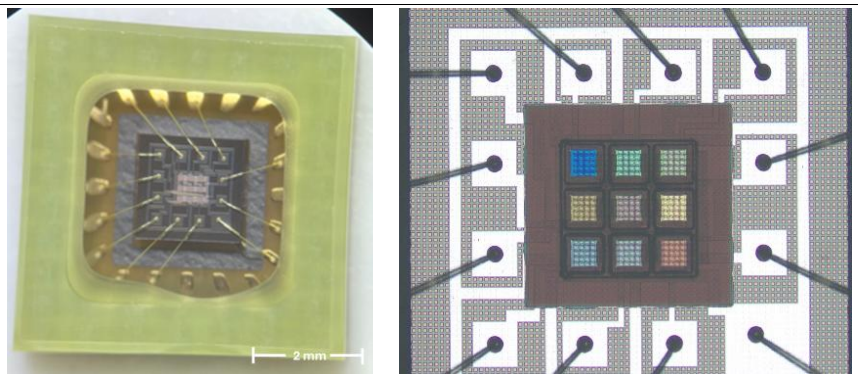

Figure 3. Packaged infrared spectroscopic sensor (left) and enlarged view of the sensor (right)

Why Global Collaboration?

Purdue University serves as a leading research hub in the state of Indiana, playing a central role in driving industrial initiatives and technology commercialization in the United States. Through key institutions such as the Birck Nanotechnology Center within Discovery Park, the Elmore Family School of Electrical and Computer Engineering, and the Discovery Park Institute, Purdue has established advanced research infrastructure and specialized facilities for optical systems and photonic engineering. These organizations provide an integrated environment for nanophotonics, sensor devices, optical system design, and performance validation, creating optimal conditions for the development and demonstration of high-performance optical technologies required in this project.

In addition, Purdue University offers strong capabilities in experimental testbed development, large-scale data handling, and industry-linked validation environments, enabling early-stage verification and systematic advancement of core technologies. These strengths support not only the reinforcement of fundamental technological competitiveness but also effective market entry strategies through close engagement with the U.S. industrial ecosystem. By leveraging Purdue's research and commercialization infrastructure, the proposed collaboration will facilitate market benchmarking, local industry engagement, and feasibility validation, positioning Purdue as a strategic gateway for global deployment and long-term competitiveness.

Synergistic Sensor-Optics Co-Design for a New Class of Industrial SWIR Scanners

This project must be pursued as a sensor–optics co-design effort that tightly integrates the Ge-based SWIR sensor platform of Korean SMEs with advanced optical engineering capabilities. Rather than treating the sensor, optics, and signal processing as independent subsystems, the proposed approach must optimize photon collection, spectral detection, and system alignment simultaneously in order to achieve reliable non-contact operation. Without this integrated design strategy, stable performance at extended standoff distances cannot be realized.

By jointly optimizing the GCMD-based SWIR sensor architecture and the high-efficiency optical collection system, the project must address the fundamental trade-offs among numerical aperture,

	field of view, depth of focus, and signal-to-noise ratio that emerge under non-contact conditions. This co-design approach is essential to ensure that optical gain is effectively converted into usable spectral information at the sensor level, rather than being degraded by misalignment, aberrations, or stray-light interference. Such system-level optimization is required to overcome the intrinsic physical limitations of distance-based spectral sensing. The outcome of this collaboration must extend beyond laboratory-level demonstrations and result in a practical, industrial-grade SWIR scanning system capable of stable and distance-invariant spectral measurement. By advancing an already commercialized sensing platform into non-contact operation, the project must establish a new technical and economic benchmark for affordable and deployable SWIR spectroscopy in industrial environments. This transition is necessary to enable large-scale adoption in automated inspection and monitoring applications. In summary, this joint R&D initiative must combine the core SWIR sensor technology of Korean SMEs with advanced optical system design and AI-based spectral compensation. Through integrated sensor–optics co-design, the project must overcome the fundamental limitations of non-contact spectral sensing and position the resulting technology as a globally competitive solution for next-generation industrial inspection and monitoring systems.
Key Words	**1. Non-contact SWIR spectroscopy** **2. Optical system optimization** **3. Distance-robust spectral sensing** **4. AI-based spectral compensation** **5. Industrial field deployment**

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (5)	**End of the Project**	Level (7)
R&D Objectives	**R&D Objectives** The primary objective of this joint R&D project is to transition a contact-based LS SWIR sensing platform into a high-performance non-contact system capable of accurate and stable spectral measurement at standoff distances of 10 cm or greater. **Detailed Objectives** To achieve this goal, the project defines the following specific R&D objectives: 1. **High-Efficiency Optical Photon Collection** 2. **Distance-Robust Signal Integrity** 3. **AI-Based Distance Compensation and Calibration** 4. **Precision Targeting and Alignment** 5. **System Integration and Industrial Demonstration**			

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R&D Target Specifications

The target performance specifications for this project are defined to bridge the gap between laboratory-scale validation and ruggedized industrial deployment. These metrics are designed to ensure that the resulting non-contact SWIR sensing system meets the operational requirements of automated, high-throughput, and optically uncontrolled environments.

Each specification is directly aligned with the project's R&D objectives and serves as a quantitative benchmark for evaluating progress from TRL 5 to TRL 7.

Table 1: Target Performance Specifications for Non-Contact SWIR Spectral Sensing System

Specifications	% of Total	Unit	Target	Note
Working Distance	30	cm	≥10	Distance from a device to a sample to achieve the target SNR above
Precision of Guidance	20	mm	≤ 5	Deviation from the sensor's field of view and the visible laser guidance
Signal-to-Noise Ratio (SNR)	20	-	≥500:1	Maximum achievable SNR at the distance of 10 cm
Classification Accuracy	30	%	≥95	Classification of two materials at the distance of 10 cm

a) **Working Distance**: Achieve a stable sensing range of ≥10 cm, significantly extending the operational envelope beyond current contact-based limitations.

b) **Precision of Guidance**: Implementation of a co-axial visible laser system to provide a parallax-free guide, ensuring the user or robotic arm is accurately aligned less than 5 mm off from the sensor's field of view at the maximum distance.

c) **Signal Integrity**: Achieve a SNR of 500:1 through periodic modulation and lock-in detection, even under signal attenuation caused by the Inverse Square Law.

d) **Classification Accuracy**: Maintain > 95% accuracy in material identification across varying distances and diverse lighting conditions, powered by distance-robust AI models.

R&D Plan

The R&D plan is structured into four tightly coupled technical workstreams that collectively enable the transition of the LS SWIR SDK from contact-based operation (TRL 5) to a ruggedized, non-contact industrial prototype (TRL 7). Each workstream is aligned with a specific technical bottleneck and is led by the partner best positioned to address it.

Phase I: Integrated Optical Design & Simulation

Phase I focuses on the design and simulation of a high-efficiency SWIR optical collection system optimized for non-contact operation at standoff distances of ≥10 cm, with emphasis on photon throughput, aberration control, and manufacturable tolerance margins.

Key Activities

High-Efficiency SWIR Collection Optics

Purdue will design a custom objective lens system—either multi-element refractive or catadioptric—optimized for the 1200–1650 nm SWIR band. The design will maximize photon throughput while maintaining spectral uniformity across the sensor surface.

- **Numerical Aperture (NA) Optimization:**
  Balance large-aperture collection with sufficient depth of field to accommodate distance variation.

- **Aberration Control:**
  Minimize chromatic and off-axis aberrations that could distort spectral signatures and degrade classification accuracy.

Stray Light and Ambient Noise Suppression
Non-contact operation introduces significant "optical pollution" from solar and industrial infrared sources. Purdue will use raytracing and Monte Carlo simulation to design physical baffles, apertures, and optical filtering strategies that isolate the target signal.

Deliverables (Phase 1)

- Optical system design files and simulation reports
- Photon collection efficiency estimates at ≥10 cm
- Tolerance budgets for mechanical integration

Phase II: Precision Aiming & Co-Axial Optical Alignment

Even millimeter-scale misalignment can result in complete signal loss due to the narrow Field of View (FoV) required for sufficient SNR.

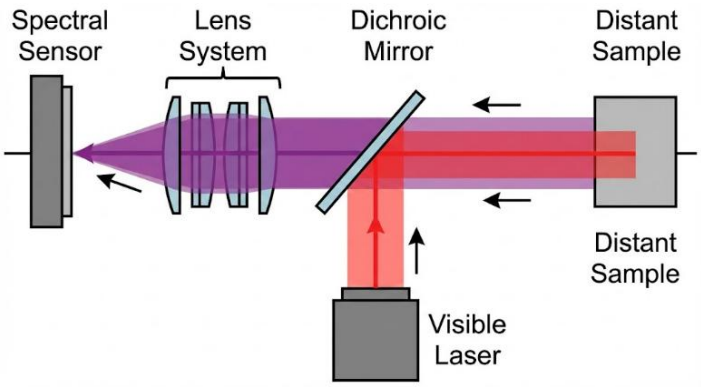

Figure 4: Simple diagram of co-axial integration of a spectral sensor and a visible laser

Key Activities

Visible Laser Targeting System
A visible laser (e.g., 635 nm or 532 nm) will be integrated to provide real-time visual guidance without interfering with SWIR measurements.

- **Spectral Isolation:**
  A SWIR band-pass filter (1200–1700 nm) ensures complete rejection of visible photons at the sensor level.

Co-Axial Optical Architecture
To eliminate parallax error, Purdue will design a co-axial configuration using a dichroic beam splitter that injects the visible laser into the same optical path as the SWIR collection optics.

- **Distance-Invariant Alignment:**
  The targeting spot remains coincidence with the sensor FoV across varying distances, enabling repeatable manual or robotic alignment.

Tolerance Analysis
Monte Carlo tolerance analysis will quantify sensitivity to decenter, tilt, and despace, ensuring manufacturability and robustness.

Deliverables (Phase 2)

- Co-axial optical layout and alignment strategy

- Laser targeting integration design
- Alignment tolerance specifications

Phase III: Distance-Robust Signal Processing & AI Compensation

Unlike conventional spectrometers, proposed GCMD-based SWIR sensor architecture is inherently suited to structured, modulated signal acquisition. This phase exploits that advantage to maintain spectral integrity under non-contact conditions.

Key Activities

Periodic Modulation & Lock-In Detection
The integrated SWIR illumination source will be periodically modulated to encode the signal in the frequency domain.

- **Frequency-Domain Extraction:**
  Using digital lock-in detection or FFT-based analysis, the system isolates the modulated spectral signal from high-intensity DC backgrounds.
- **Ambient Noise Rejection:**
  Non-periodic interference from ambient lighting is effectively suppressed, preserving SNR at extended distances.

Distance-Invariant Feature Extraction
To avoid the cost and complexity of Time-of-Flight (ToF) hardware, algorithmic distance compensation within the Infrared-AI™ engine will be implemented.

- **Relative Spectral Normalization:**
  Techniques such as Standard Normal Variate (SNV) and Multiplicative Scatter Correction (MSC) emphasize spectral shape over absolute intensity.
- **AI-Based Distance Inference:**
  Machine learning models infer distance-related attenuation from spectral degradation patterns and apply real-time gain correction.

Deliverables (Phase 3)

- Distance-robust signal processing pipeline
- AI models validated across multiple distances and lighting conditions
- Performance benchmarks for SNR and classification accuracy

Phase IV: System Integration, Prototyping & Field Validation

The final phase integrates Purdue's optical system with Korean SMEs' sensor, electronics, and software into a ruggedized prototype suitable for industrial validation.

Key Activities

Precision Mechanical Integration
Given the small active area of the sensor, mechanical stability is critical. CNC-machined monolithic lens barrels with thermally matched materials and active alignment will be employed during assembly.

Hardware–Software Co-Optimization
An iterative feedback loop will refine optical parameters and AI models jointly, ensuring optimal performance with sub-100 ms latency suitable for edge deployment.

Field Demonstration
The prototype will be tested on representative materials (plastics, textiles, organic matter) under dynamic conditions such as moving conveyor belts and temperature ranges of 0–40 °C.

Deliverables (Phase 4)

	• Ruggedized non-contact SWIR prototype (TRL 7) • Field validation data and reliability test results • Demonstration-ready system for industrial stakeholders

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Expected R&R of Research Institute (Purdue University)** Purdue University will be responsible for the optical system design and theoretical performance optimization required to enable reliable non-contact SWIR spectral sensing. This includes end-to-end optical path modeling, high-efficiency photon collection system design, and simulation-driven optimization to counteract distance-induced signal attenuation. Specifically, Purdue University will lead: • Design of high–numerical aperture SWIR optical architectures optimized for the 1200–1650 nm wavelength range • Aberration correction, stray-light suppression, and tolerance-aware optical optimization • Co-axial optical alignment design integrating visible laser targeting with SWIR collection optics • Simulation-based validation of optical performance, including ray tracing and Monte Carlo tolerance analysis • Technical guidance on manufacturability and alignment constraints to support system integration Through these activities, Purdue University will provide the foundational optical engineering required to translate 'LS SWIR SDK sensor platform into a viable non-contact system.
Expected R&R of Korean SME Partner	**Expected R&R of Korean SME Partner** The Korean SME partner will lead the overall system development, integration, and validation of the non-contact SWIR sensing platform. Building on its proprietary Ge-based GCMD sensor technology and commercialized LS SWIR SDK, the company will be responsible for transforming the optical designs into a functional, ruggedized industrial prototype. Specifically, the Korean SME partner will lead: • Development and fabrication of LS SWIR SDK sensor modules and supporting electronics • Implementation of modulated illumination, lock-in detection, and distance-robust signal acquisition firmware • Development of Infrared-AI™–based calibration, normalization, and classification algorithms • Precision mechanical integration, active alignment, and prototype assembly • System-level validation, industrial field testing, and performance benchmarking These responsibilities will be supported by the company's in-house nanofabrication and prototyping facilities, which enable rapid prototyping and iterative hardware validation, as well as access to

	advanced external fabrication infrastructure and process expertise. The Korean SME partner will also be responsible for defining industrial use cases, engaging end users, and establishing commercialization pathways for the resulting technology.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project aims to deliver a high-performance non-contact SWIR spectral sensing system that expands the industrial applicability and commercial value of the Korean SMEs' LS SWIR SDK platform. By enabling reliable spectral measurement at standoff distances of 10 cm or greater, the system will support deployment in automated inspection, recycling, environmental monitoring, and robotic sensing environments. **(1) Technical and Industrial Outcomes** The project will upgrade the existing contact-based sensing platform into a distance-robust, non-contact industrial system capable of operating under real-world conditions. This advancement builds on an already commercialized platform, enabling scalable deployment without major hardware redesign. As a result, current users will benefit from improved system flexibility and higher throughput. **(2) Market Expansion and Commercial Impact** The non-contact capability will enable Korean SMEs to enter new industrial markets such as conveyor-based sorting, robotic inspection, and continuous monitoring systems. By leveraging an existing platform, the project shortens the commercialization cycle and accelerates time-to-market, creating new revenue opportunities. **(3) Validation Through Prior Research and Commercialization** The feasibility of compact spectral sensing has been demonstrated through prior commercialization and applied research in visible and near-infrared applications [2-6]. These experiences validate the underlying sensing approach and support the technical credibility of the proposed SWIR-based non-contact system. **(4) International Collaboration and Ecosystem Impact** Collaboration with Purdue University will enhance optical design capabilities and strengthen U.S.–Korea R&D cooperation. The project will promote knowledge exchange, researcher training, and long-term collaboration, contributing to the development of globally competitive talent and sustainable innovation partnerships.

6. References

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